

Face Detection and Recognition of 37 Students using MediaPipe and ArcFace

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Abstract

This project presents a face detection and recognition system using the MediaPipe framework for detection and ArcFace for recognition. The system is designed to detect and recognize the faces of 37 individuals from a classroom dataset, capturing facial images from various positions and angles to simulate real-world variability. MediaPipe provides real-time face detection and landmark extraction, making it suitable for lightweight applications on CPUs and mobile devices. Following detection, ArcFace is applied to generate highly discriminative facial embeddings for robust recognition. The experimental results demonstrate high recognition accuracy under diverse lighting conditions, facial orientations, and head poses. This system demonstrates strong potential for deployment in automated attendance tracking, access control, and intelligent classroom environments.

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1 Introduction

Face detection and recognition have become central components in modern computer vision applications, with widespread use in surveillance, biometric authentication, smart classrooms, and human-computer interaction. These technologies aim to identify and analyze human faces within images or video streams, providing the foundation for systems such as automated attendance tracking, access control, and emotion recognition.

Traditional face detection methods, such as Haar cascades and HOG-based detectors, have been limited in accuracy and robustness, particularly under challenging conditions such as varying illumination, head poses, and occlusions. With the rise of deep learning, frameworks such as MediaPipe have significantly improved real-time performance for face detection and landmark localization.

At the same time, embedding-based recognition models such as ArcFace have revolutionized face recognition, achieving near-human performance by mapping facial features into a highly discriminative embedding space.

This project focuses on implementing a face detection and recognition system where MediaPipe is used for real-time face detection and ArcFace is employed for recognition. The system is tested on a custom classroom dataset containing 37 individuals, with images captured from different angles and lighting conditions to emulate real-world variability.

2 Literature Review

Face detection and recognition are critical tasks in computer vision, with applications in surveillance, access control, human-computer interaction, and automated attendance systems. Traditional methods for face detection include techniques such as Haar cascades (Viola-Jones algorithm) and Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM). Although these methods were widely used because of their simplicity and speed, they often struggle under varying lighting conditions, occlusions, or different facial orientations.

With the rise of deep learning, convolutional neural networks (CNNs) have significantly improved face detection and recognition accuracy. Models like MTCNN (Multi-task Cascaded Convolutional Networks) [2], RetinaFace, and Dlib-based detectors have been effective in handling complex variations in real-world datasets. For recognition, embedding-based models such as FaceNet [1], DeepFace, and ArcFace [3] have achieved near-human performance by mapping facial features into a high-dimensional vector space.

Google’s MediaPipe framework has emerged as an efficient and lightweight solution for real-time vision tasks. It provides pre-trained pipelines for face detection and face mesh extraction, optimized for cross-platform deployment on mobile, web, and embedded devices [4]. Unlike heavy models requiring GPUs, MediaPipe achieves real-time performance on CPUs, making it suitable for classroom deployment and attendance systems.

This project leverages MediaPipe’s efficient detection pipeline along with ArcFace’s state-of-the-art recognition capabilities, combining lightweight detection with high-accuracy recognition.

3 System Overview

The system integrates two major components:

- **MediaPipe for Detection:** Used to localize faces and extract bounding boxes in real-time.
- **ArcFace for Recognition:** Generates embeddings that represent unique facial features, which are then compared using cosine similarity to classify identities.

4 Methodology

4.1 Dataset Preparation

Images of 37 students were collected under varied lighting and pose conditions. The dataset was split into training and testing subsets.

4.2 Face Detection with MediaPipe

MediaPipe’s Face Detection module was applied to extract bounding boxes of faces. Detected regions of interest were cropped and aligned for further processing.

4.3 Face Alignment and Preprocessing

Using MediaPipe’s Face Mesh, key facial landmarks were extracted to normalize poses and improve robustness to orientation variations. Cropped faces were resized and normalized to the input size required by ArcFace.

4.4 Face Recognition with ArcFace

ArcFace was used to generate 512-dimensional embeddings from the preprocessed face crops. These embeddings were compared with a database of enrolled students using cosine similarity. The nearest embedding below a threshold was assigned as the recognized identity.

4.5 System Pipeline

The complete system pipeline consisted of:

1. Input image/video stream.
2. MediaPipe face detection.
3. Face alignment and preprocessing.
4. ArcFace embedding extraction.
5. Identity classification using cosine similarity.

5 Results and Evaluation

The system was evaluated on recognition accuracy, detection speed, and robustness under varying conditions. MediaPipe achieved real-time detection speeds, while ArcFace provided highly discriminative recognition results. The combination achieved strong performance even under non-ideal lighting and varied head poses.

6 Discussion

The results indicate that MediaPipe is effective for lightweight detection, while ArcFace ensures high recognition accuracy. Unlike YOLO-based systems, this combination balances efficiency and accuracy, allowing deployment on CPU-only devices such as classroom PCs.

7 Conclusion

This project demonstrates a MediaPipe + ArcFace based face detection and recognition system for classroom environments. The framework provides real-time detection and reliable recognition, making it a strong candidate for automated attendance and access control.

8 Future Work

Future improvements may include:

- Expanding the dataset to include larger classes and varying demographics.
- Integration with cloud-based attendance systems.
- Incorporating temporal information from video streams for improved recognition stability.

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